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Учебник

Как настроить частную виртуальную машину и хост-бастион с помощью облачных брандмауэров

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Частные Капли

Капли

Создание сетей

Безопасность



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How to Set Up a Private Droplet

✓ Generally Available

VPC-only compute with no public IP



Содержание



Популярные темы



Частная виртуальная машина не имеет публичного сетевого интерфейса, поэтому вы не можете подключиться к ней по SSH напрямую. Стандартный способ подключения — через **хост-бастион** (хост-промежуток): небольшую виртуальную машину с публичным IP-адресом в той же **виртуальной частной сети**, к которой вы подключаетесь *через*. Чтобы обеспечить безопасность этого пути, на каждую виртуальную машину устанавливается **облачный брандмауэр**: один позволяет подключаться только к хосту-бастиону, а другой — только к частной виртуальной машине.

В этом уроке мы с нуля разбираем все четыре части:

1. Создайте частную виртуальную машину.
2. Создайте хост-бастион в том же VPC.
3. Настройте облачный брандмауэр на бастионе (только для SSH с вашего IP).
4. Настройте облачный брандмауэр на частной виртуальной машине (только для SSH с бастиона).

После этого выполните действия, описанные в разделе **Как подключиться к частной виртуальной машине**.

Что на самом деле представляет собой Private Droplet

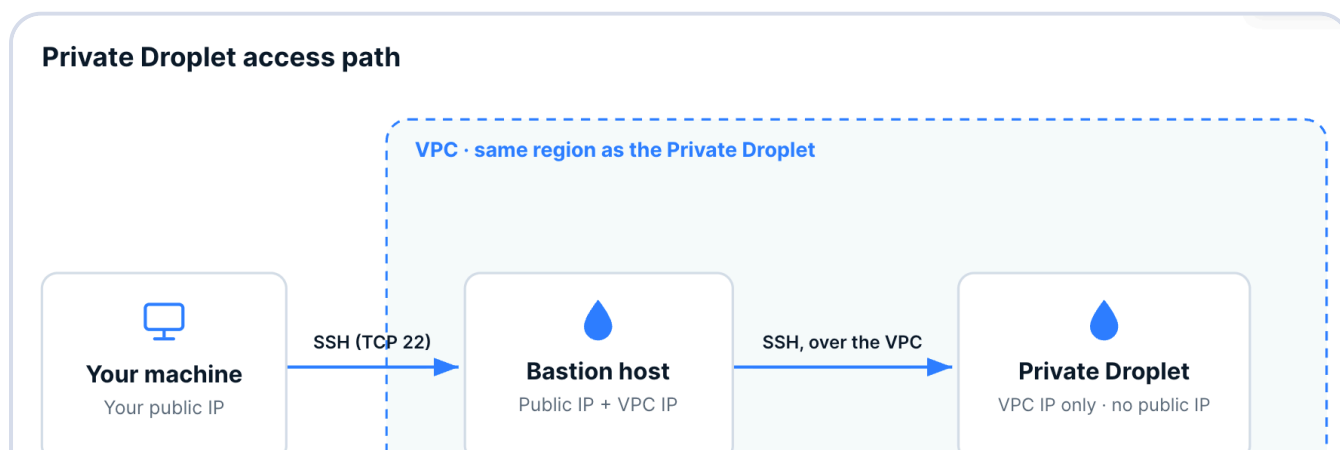


Частная виртуальная машина имеет один частный сетевой интерфейс (eth0) и не имеет публичного IPv4. Традиционная виртуальная машина имеет два интерфейса: публичный `eth0` и частный `eth1`. Разница носит структурный характер, а не является параметром, который можно изменить постфактум.

- Нет публичного входящего трафика, кроме как через балансировщик нагрузки, подключенный к VPC.
- Нет исходящего интернет-трафика, если только VPC не направляет его через шлюз NAT.
- Используется локальный DNS-преобразователь VPC.
- Невозможно преобразовать в традиционную виртуальную машину или обратно, приватный статус фиксируется при создании. Чтобы «преобразовать» виртуальную машину, нужно сделать ее снимок и создать заново.
- Нет дополнительной платы: приватная виртуальная машина стоит столько же, сколько стандартная виртуальная машина того же размера.

Принцип проектирования, который необходимо усвоить, заключается в том, что **частное остается частным**. Нет возможности подключить публичный IP-адрес, зарезервированный IP-адрес или адрес BYOIP позже. Эта неизменность и является отличительной особенностью. Она позволяет аудитору обосновать требование об изоляции.

Для простоты понимания приведем схему пути доступа к Private Droplet:



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Когда следует использовать Private Droplet



Отличительной особенностью Private Droplet является отсутствие публичного сетевого интерфейса. К ней нельзя получить прямой доступ из интернета, и нет возможности добавить к ней публичный IP-адрес. Эта изолированность — причина, по которой стоит выбрать Private Droplet, а также источник всех компромиссов, описанных ниже.

Хорошо подходит:

- **Внутренние уровни, которые никогда не должны быть общедоступными** — базы данных, внутренние API, серверы моделей и другие сервисы, доступ к которым должен быть только у других ресурсов внутри вашего VPC.
- **Многоуровневые архитектуры**, в которых доступен только внешний уровень (за балансировщиком нагрузки), а все, что находится за ним, остается закрытым.
- **Рабочие нагрузки со строгими требованиями к сетевой изоляции** — например, команды, работающие над соблюдением требований HIPAA, SOC 2 или PCI, где требование звучит так: «У этой системы нет публичного IP-адреса», а не «У этой системы есть публичный IP-адрес, который мы изолируем брандмауэром». Структурную гарантию проще подтвердить при аудите, чем правило, которое может быть изменено. *(Уточните формулировки, связанные с соблюдением нормативных требований, у своей команды по соблюдению нормативных требований — отсутствие публичного интерфейса — это факт, а не сертификация.)*

When a Private Droplet is the wrong tool:

- **A simple public-facing server.** If a service needs to accept traffic from the internet directly, a standard Droplet with a Cloud Firewall is simpler and does the job.
- **Workloads that need constant, unpredictable outbound access.** Private Droplets reach the internet only through a NAT gateway, which adds cost and a moving part. If a box needs broad egress all the time, weigh whether the isolation is worth it.
- **Quick experiments or demos,** where the operational overhead of a bastion and NAT gateway isn't justified.

Consequences to plan for:

- **No direct SSH.** You reach a Private Droplet only through a bastion host, which is exactly what this tutorial sets up. There is also no web-based Droplet console path that bypasses this; plan your access model before you need it.
- **No outbound internet by default.** Package updates, container pulls, and downloads require a [VPC NAT gateway](#) set as the VPC's default route. (A NAT gateway provides outbound only — it does not accept inbound connections, so it cannot replace the bastion.)
- **The networking type is permanent.** A Private Droplet cannot be converted to a public Droplet, and an existing Droplet cannot be converted to private. To change the type, snapshot the workload and recreate it.
- **No cost premium, but added operational pieces.** A Private Droplet costs the same as a standard Droplet of the same size. The real cost is operational: you take on a bastion host and, if you need egress, a NAT gateway.
- **Some integrations differ.** As of the current docs, Private Droplets can't be created from the Control Panel, custom images aren't supported, and DOKS worker nodes aren't Private Droplets. Check the [Private Droplets documentation](#) for the current list before you design around them.

Prerequisites



- A DigitalOcean account and a [personal access token](#).
- An SSH key added to your account (you'll reference it by fingerprint). The same key should be installed on both Droplets. See [How to Add SSH Keys to Droplets](#).
- A VPC in your target region (the region's default VPC is fine). Note its name and region.
- `doctl` installed and authenticated (`doctl auth init`). Upgrade `doctl` if `--enable-public-networking` is missing from `doctl compute droplet create --help` .

- Your own current public IP address, for locking down SSH to the bastion. You can find it by visiting any “what is my IP” service.

Step 1 — Create the Private Droplet ^

A Private Droplet is created by turning public networking **off** at creation time. This is permanent, you can’t add public networking later, and you can’t convert an existing Droplet into a Private Droplet.

Create it with `doctl`, placing it in your chosen VPC:

Copy

```
doctl compute droplet create private-app-01 \
  --region sfo3 \
  --size s-1vcpu-1gb \
  --image ubuntu-24-04-x64 \
  --vpc-uuid <your-vpc-uuid> \
  --ssh-keys <your-ssh-key-fingerprint> \
  --enable-public-networking=false
```

The `--enable-public-networking=false` flag is what makes the Droplet private (equivalent to `"public_networking": false` in the API).

After it’s created, record two values you’ll need later:

- The **Droplet ID** — run `doctl compute droplet list --format ID,Name,PublicIPv4,PrivateIPv4` .
- The **VPC IP address** — shown in the Control Panel under **Connection Details**, or retrievable from the bastion later via the metadata path `interfaces/private/0/ipv4/address` . (A Private Droplet has only a VPC IP, no public IP.)

You can also create the Private Droplet using the Control Panel.

1. Go to **Droplets > Create Droplet**.
2. Choose the **region** you want to create the Private Droplet in.
3. Pick the **size** you want to create the Private Droplet in.

4. In the Networking section, simply uncheck the **Public IPv4 address** checkbox.

This will create a Private Droplet in the same VPC and region as the one you selected.

The screenshot shows the DigitalOcean Droplet creation interface. The 'SSH Keys' tab is active, showing a list of keys: 'james-key', 'Melani', and 'Anish' (selected). The 'Networking' section is visible, showing the 'Public IPv4 address' checkbox is unchecked. A blue box highlights the information message: 'This Droplet will be private'. The 'Summary' section on the right shows the 'Compute' plan details and the 'Total cost' of \$32.00/month. The 'Create Droplet' button is visible.

SSH Keys Password

Add an SSH Key for authentication
An SSH key pair is a more secure way to connect to your Droplet

[Add SSH Key](#)

☐ Select All 1 / 3

☐ james-key

☐ Melani

☒ Anish

Networking
Public IP addresses are optional. A private IPv4 address is always assigned for secure internal traffic within the same VPC.

☐ **Public IPv4 address**
Allows your Droplet to send and receive traffic from the public internet.
Disable to restrict this Droplet to your VPC.

☐ **Public IPv6 address**
Built-in authentication and privacy support, and makes it difficult for hackers to launch scanning attacks.

This Droplet will be private
This Droplet won't have public IPs now or later, and can only be reached through a resource with public access in the same VPC.
Add outbound internet access later with a [NAT Gateway](#), or route public traffic to it with a [Load Balancer](#).

Summary

Compute \$32.00/mo
Plan Type: Basic
CPU Option: Premium
Intel
vCPU: 2
RAM: 4 GB
Disk: 120 GB
Bandwidth: 4 TB
Slug: s-2vcpu-4gb-120gb-intel

Total cost \$32.00/month
\$0.048/hour

[Create Droplet](#)

Want to maximize efficiency and cost?
Programmatically manage Droplets in a repeatable and re-usable way with our API. [Create via API](#)

Next, you will create the bastion host.

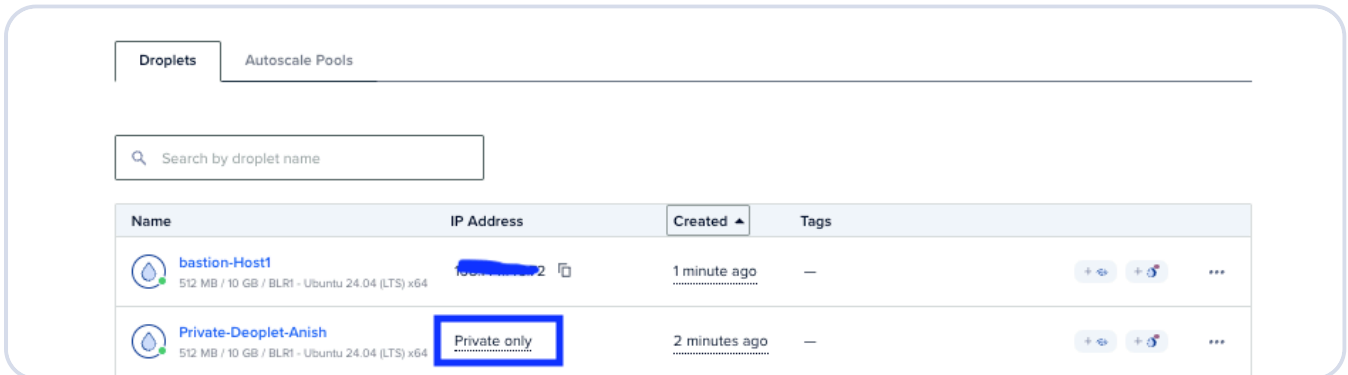
Step 2 — Create the bastion host

The bastion is an ordinary Droplet with a public IP. The only hard requirement is that it sits in the **same VPC and region** as the Private Droplet, so it can reach it over the private network.

Using the Control Panel:

1. Click **Create > Droplets**.
2. Choose the **same region** as your Private Droplet.
3. Pick a small size (for example, the smallest shared-CPU plan, a bastion does no real work).
4. Under **VPC Network**, select the **same VPC** as your Private Droplet. This is the critical setting; if the bastion is in a different VPC, it won't be able to reach the Private Droplet.

5. Under **Authentication**, select the **same SSH key** you used for the Private Droplet (or another key you trust for bastion access).
6. Name it something recognizable, such as `bastion-Host1` , and click **Create Droplet**.



When it finishes, note the bastion's **public IP** and its **Droplet ID** (visible in the Droplet list).

doctl equivalent: `doctl compute droplet create bastion-01 --region sfo3 --size s-1vcpu-1gb --image ubuntu-24-04-x64 --vpc-uuid <same-vpc-uuid> --ssh-keys <your-ssh-key-fingerprint> — note there's no -enable-public-networking=false here, because the bastion needs its public IP.`

Step 3 — Add a Cloud Firewall to the bastion

DigitalOcean Cloud Firewalls deny everything that isn't explicitly allowed: if there are no inbound rules, no inbound traffic is permitted. They are also **stateful**, so return traffic for an allowed connection is permitted automatically — you don't need a matching outbound rule for SSH replies. (Cloud Firewalls are separate from any on-Droplet firewall like `ufw`.)

For the bastion, the only inbound you want is SSH from your own IP.

Using the Control Panel:

1. Go to **Networking > Firewalls > Create Firewall**.
2. Name it `bastion-ssh` .
3. Under **Inbound Rules**, keep the **SSH (TCP 22)** rule but change its **Sources** from the default "All IPv4/All IPv6" to **your public IP** (for example `203.0.113.10/32`). Remove the open defaults.

4. Leave the **Outbound Rules** at their defaults (all TCP, all UDP, ICMP), so the bastion can still reach updates and the VPC.
5. Under **Apply to Droplets**, select your bastion (`bastion-Host1`).
6. Click **Create Firewall**.

Note: On the bastion's firewall (Firewall A): enter your own personal computer's public IP. This is the only place your laptop's IP goes. You're the one connecting to the bastion from the outside, so the bastion should accept SSH only from you.

If you are using a mac you can use the below command to get your public/external IP address:

Copy

```
curl ifconfig.me
```

Add inbound rules

Inbound firewall rules define the traffic allowed to your Droplets on which ports and from which sources. All other incoming traffic is blocked.

Type	Protocol	Port range	Sources
SSH	TCP	22	 ...

+ Add another

Add outbound rules

Outbound firewall rules define the traffic allowed to leave your Droplets on which ports and to which destinations. All other outbound traffic is blocked.

Type	Protocol	Port range	Destinations
ICMP	ICMP		 All IPv4  All IPv6 ...
All TCP	TCP	All ports	 All IPv4  All IPv6 ...
All UDP	UDP	All ports	 All IPv4  All IPv6 ...

+ Add another

Apply to Droplets

Select Droplets to apply your firewalls to. You can add Droplets individually by searching their name, or in bulk by selecting a tag shared across multiple Droplets.

 bastion-Host1 x

Search for a Droplet or a tag

Choose a name

Names can only contain alphanumeric characters, dashes, and periods.

Create Firewall

doctl equivalent:

Copy

```
doctl compute firewall create \  
  --name "bastion-ssh" \  
  --inbound-rules "protocol:tcp,ports:22,address:<your-ip>/32" \  
  --outbound-rules "protocol:tcp,ports:all,address:0.0.0.0/0 pro  
  --droplet-ids <bastion-droplet-id>
```

Step 4 — Add a Cloud Firewall to the Private Droplet



For the Private Droplet, the only inbound you want is SSH **from the bastion**.

Using the Control Panel:

1. Go to **Networking > Firewalls > Create Firewall**.
2. Name it `private-droplet-ssh` .
3. Under **Inbound Rules**, keep **SSH (TCP 22)** and set its **Sources** to your **bastion Droplet**: start typing its name (`bastion-01`) and select it. This restricts SSH to traffic coming from the bastion over the VPC.
4. Leave the **Outbound Rules** at their defaults.
5. Under **Apply to Droplets**, select your Private Droplet (`private-app-01`).
6. Click **Create Firewall**.

Add inbound rules

Inbound firewall rules define the traffic allowed to your Droplets on which ports and from which sources. All other incoming traffic is blocked.

Type	Protocol	Port range	Sources
SSH	TCP	22	 bastion-Host1 ...

[+ Add another](#)

Add outbound rules

Outbound firewall rules define the traffic allowed to leave your Droplets on which ports and to which destinations. All other outbound traffic is blocked.

Type	Protocol	Port range	Destinations
ICMP	ICMP		 All IPv4  All IPv6 ...
All TCP	TCP	All ports	 All IPv4  All IPv6 ...
All UDP	UDP	All ports	 All IPv4  All IPv6 ...

[+ Add another](#)

Apply to Droplets

Select Droplets to apply your firewalls to. You can add Droplets individually by searching their name, or in bulk by selecting a tag shared across multiple Droplets.

 Private-Droplet-Anish 

Search for a Droplet or a tag

Choose a name

Names can only contain alphanumeric characters, dashes, and periods.

private-droplet-ssh

Create Firewall

Both Droplets now exist, they share a VPC, and the firewalls allow exactly one path in: your machine to the bastion, and the bastion to the Private Droplet. From here, follow the SSH steps in the official guide:

You can follow the steps in [How to Connect to a Private Droplet](#) to connect to the Private Droplet.

In short, you'll use `ssh -J` (ProxyJump) to hop through the bastion to the Private Droplet's VPC IP. Remember the default login user on most DigitalOcean images is `root`.

Copy

```
ssh -J <bastion-user>@<bastion-public-ip> <private-user>@<private-vpc-ip>
```

For example:

Copy

```
ssh -J root@123.456.789.012 root@10.116.0.2
```

You can find the **Private Droplet's VPC IP** in the Control Panel under **Networking > VPC**.

← Back to VPC

 **default-blr1**

BLR1 / 3 Resources / 10.122.0.0/20

Resources

Peering Connections

NAT Gateways

Settings

NETWORK DETAILS

IP Range

10.122.0.0/20

Available Addresses

4,087

RESOURCES (3)

2 Droplets

1 Database

Droplets (2)

Name	Private IP	Public IP	
 bastion-Host1	 10.122.0.3	168.144.146.72	...
 Private-Droplet-Anish	 10.122.0.2	None	...

And thats it! You have successfully created a Private Droplet and a bastion host with Cloud Firewalls. You can now connect to the Private Droplet using the bastion host as a jump host.

Quick Verification Checklist

Make sure the following are true:

- Both Droplets show the **same VPC** under **Networking > VPC**.
- The bastion firewall allows **TCP 22 from your IP only** — not “All IPv4”.
- The Private Droplet firewall allows **TCP 22 from the bastion** (or the VPC CIDR) and nothing else inbound.
- Each firewall is actually **applied** to the right Droplet (check **Apply to Droplets**).

Notes and limitations

- A NAT gateway is only needed if the Private Droplet requires **outbound** internet (package updates, image pulls). It does **not** provide inbound access, so you can't SSH through it — the bastion is required. See [How to Create a NAT Gateway](#).
- Cloud Firewalls are stateful and are separate from any on-Droplet firewall (`ufw` , `firewalld`); if you run both, make sure their rules don't conflict.
- For other Private Droplet behavior and current limitations, see the [Private Droplets documentation](#).

Further reading

- [How to Connect to a Private Droplet](#)
- [Private Droplets — overview and limitations](#)
- [Cloud Firewalls — how to configure rules](#)
- [VPC best practices \(bastion guidance\)](#)

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Anish is a Sr Technical Content Strategist and Team Lead at DigitalOcean with 7+ years of experience as an SRE at Nutanix and Cloud consultant at AMEX, and technical writing at DOCN, and shipping deep infra and AI inference tutorials that help developers deploy production-ready applications on DigitalOcean.

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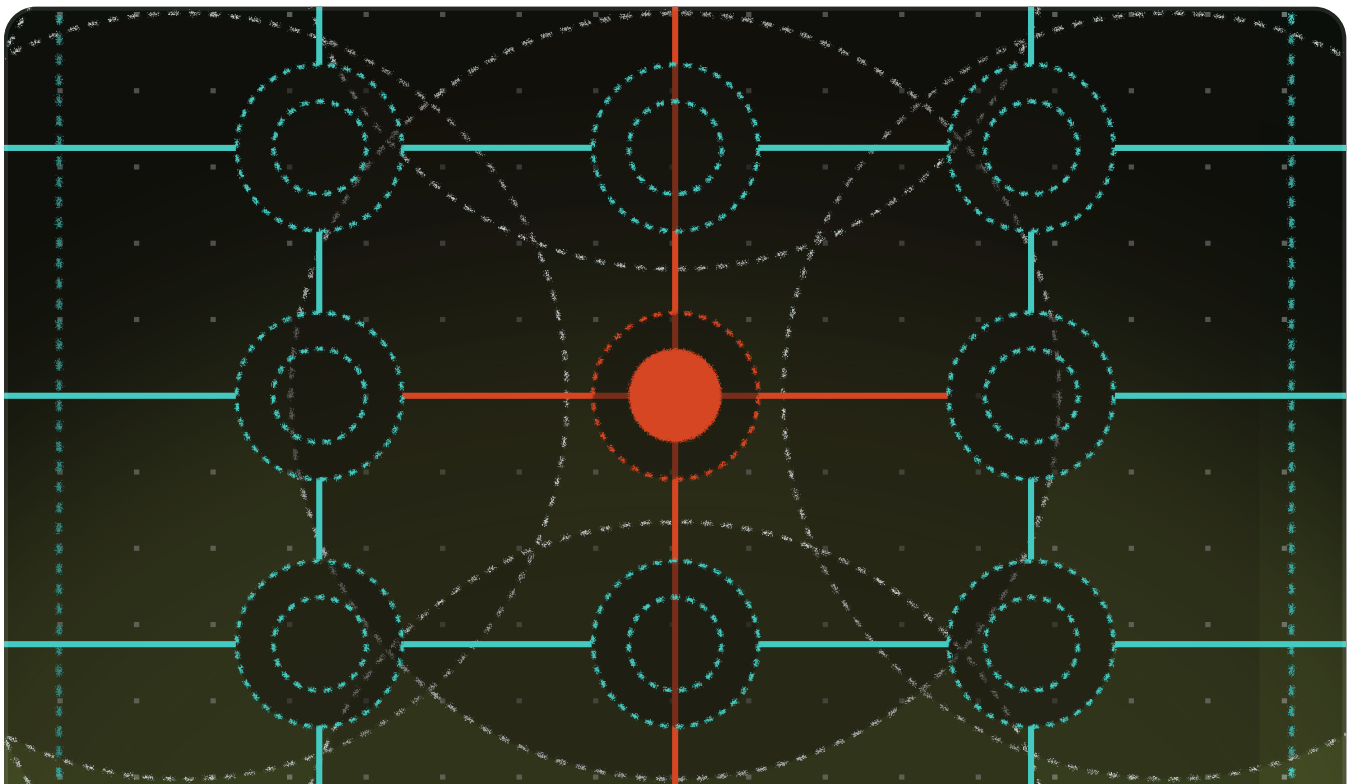
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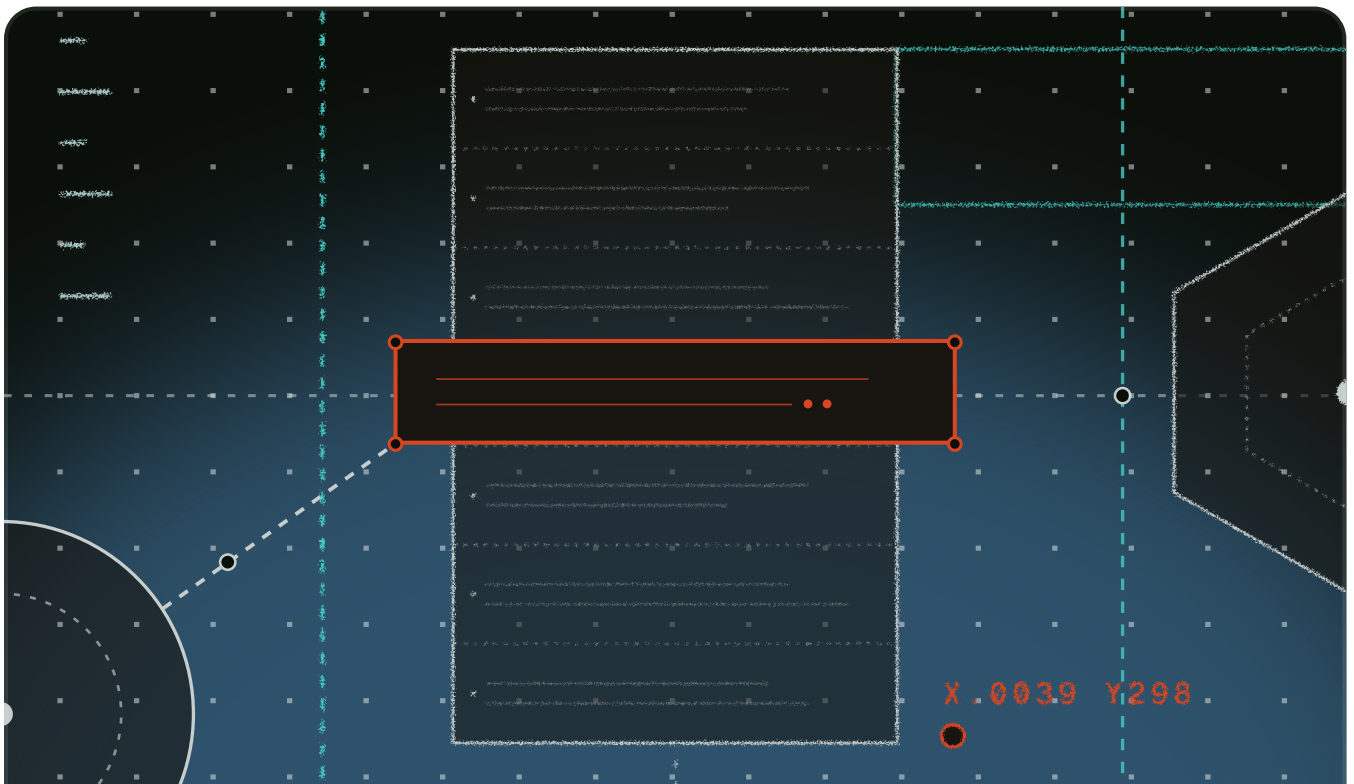
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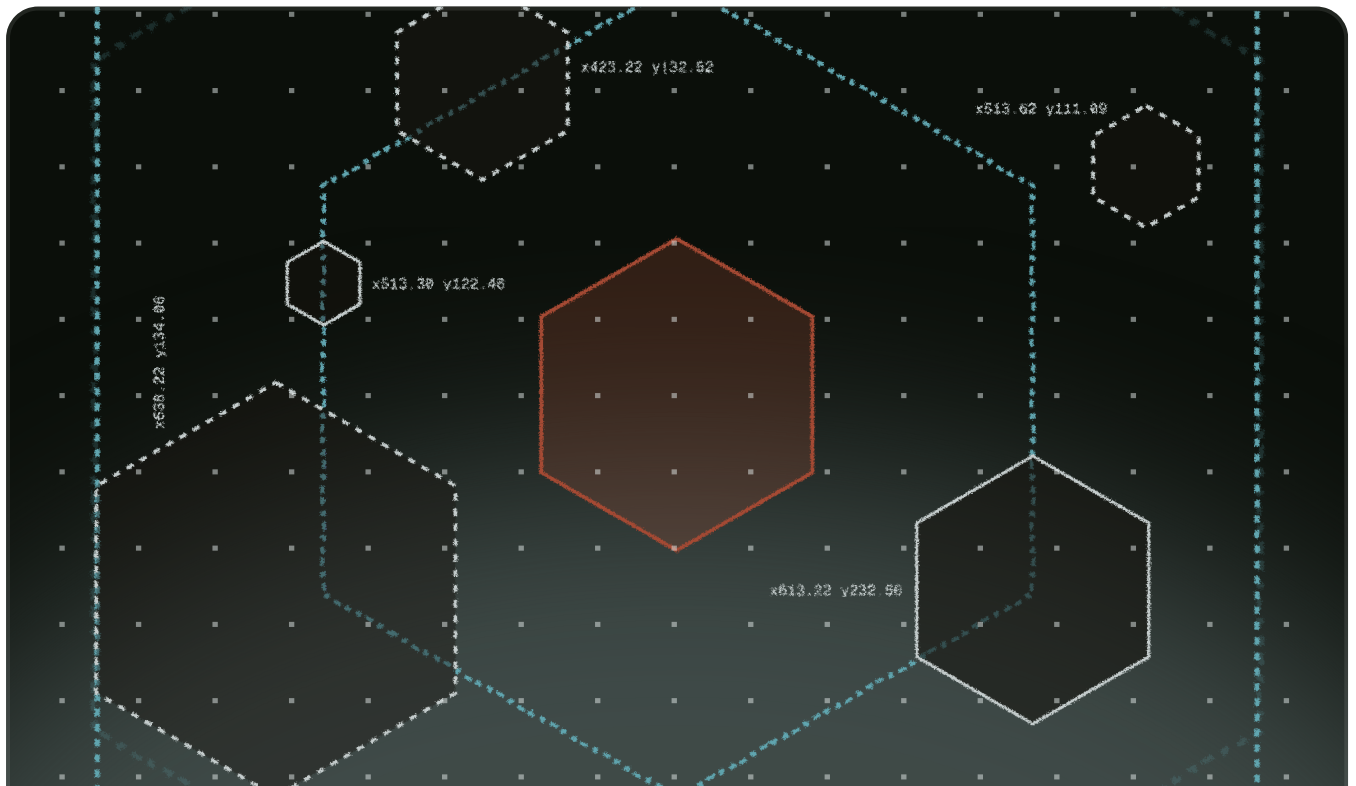
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